

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 7: Coordinate Geometry

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 The distance formula

1. The centre of a circle is $(x - 7, 2x)$. The circle passes through $(-9, 11)$ and its radius is $5\sqrt{2}$ units. Find the possible values of x . **CBSE 2026, 2 marks**

SOLUTION

The distance from the centre to $(-9, 11)$ equals the radius.

$$(x - 7 + 9)^2 + (2x - 11)^2 = (5\sqrt{2})^2$$

$$x^2 + 4x + 4 + 4x^2 - 44x + 121 = 50, \text{ so } 5x^2 - 40x + 75 = 0$$

$$x^2 - 8x + 15 = 0, \text{ so } (x - 3)(x - 5) = 0$$

Answer: $x = 3$ or $x = 5$

2. A circle has its centre at $(2a, a - 7)$ and passes through the point $(11, -9)$. Its diameter is $10\sqrt{2}$ units. Find the values of a . **CBSE 2025, 2 marks**

SOLUTION

Radius = $\frac{10\sqrt{2}}{2} = 5\sqrt{2}$ units, so the distance from the centre to $(11, -9)$ is $5\sqrt{2}$.

$$(2a - 11)^2 + (a - 7 + 9)^2 = (5\sqrt{2})^2$$

$$4a^2 - 44a + 121 + a^2 + 4a + 4 = 50, \text{ so } 5a^2 - 40a + 75 = 0$$

$$a^2 - 8a + 15 = 0, \text{ so } (a - 3)(a - 5) = 0$$

Answer: $a = 3$ or $a = 5$

3. $\triangle ABC$ has vertices $A(5, -6)$, $B(6, 4)$ and $C(0, 0)$. Find the length of its median AD . (a) $\sqrt{68}$ units (b) $2\sqrt{15}$ units (c) $\sqrt{101}$ units (d) 10 units **CBSE 2024, 1 mark**

SOLUTION

D is the mid-point of BC : $D = \left(\frac{6+0}{2}, \frac{4+0}{2}\right) = (3, 2)$

$$AD = \sqrt{(5-3)^2 + (-6-2)^2} = \sqrt{4+64} = \sqrt{68}$$

Answer: (a)

4. The points $(3, -5)$ and $(x, -5)$ are 15 units apart. Find the possible values of x . (a) 12, -18 (b) -12, 18 (c) 18, 5 (d) -9, -12 **CBSE 2024, 1 mark**

SOLUTION

Both points have the same y -coordinate, so the distance is $|x - 3| = 15$.

$x - 3 = 15$ or $x - 3 = -15$, so $x = 18$ or $x = -12$.

Answer: (b)

5. Find the distance between $(0, 2\sqrt{5})$ and $(-2\sqrt{5}, 0)$. (a) $2\sqrt{10}$ units (b) $4\sqrt{10}$ units (c) $2\sqrt{20}$ units (d) 0 **CBSE 2023, 1 mark**

SOLUTION

Distance = $\sqrt{(-2\sqrt{5} - 0)^2 + (0 - 2\sqrt{5})^2} = \sqrt{20 + 20} = \sqrt{40} = 2\sqrt{10}$ units

Answer: (a)

6. Two vertices of an equilateral triangle are $(-5, 3)$ and $(5, 3)$, and the origin lies inside the triangle. Find the third vertex. Take $\sqrt{3} = 1.7$. **CBSE 2023, 3 marks**

SOLUTION

The side joining the two given vertices has length 10, and its mid-point is $(0, 3)$.

The third vertex is equidistant from the two given points, so it lies on the y -axis. Let it be $(0, y)$.

$(0 - 5)^2 + (y - 3)^2 = 10^2$, so $(y - 3)^2 = 75$ and $y = 3 \pm 5\sqrt{3} = 3 \pm 8.5$

$y = 11.5$ puts the whole triangle above the line $y = 3$, so the origin would be outside it. Hence $y = -5.5$.

Answer: $(0, -5.5)$

7. Assertion (A): $P(0, 2)$ is the point where the line $3x + 2y = 4$ meets the y -axis. Reason (R): The distance of $P(0, 2)$ from the x -axis is 2 units. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true **CBSE 2023, 1 mark**

SOLUTION

On the y -axis $x = 0$: $2y = 4$, so $y = 2$. The line meets the y -axis at $(0, 2)$, so A is true.

The distance of a point from the x -axis is its y -coordinate, here 2 units, so R is true. But R does not tell us why P lies on the line.

Answer: (b)

8. Find the point on the y -axis that is equidistant from $(5, -2)$ and $(-3, 2)$. **CBSE 2019, 3 marks**

SOLUTION

Let the point be $P(0, y)$. Equate the squares of the two distances.

$(0 - 5)^2 + (y + 2)^2 = (0 + 3)^2 + (y - 2)^2$

$25 + y^2 + 4y + 4 = 9 + y^2 - 4y + 4$, so $8y = -16$ and $y = -2$.

Answer: $(0, -2)$

9. Write the distance of the point $P(x, y)$ from the origin. **CBSE 2018, 1 mark**

SOLUTION

$$\text{Distance} = \sqrt{(x-0)^2 + (y-0)^2}$$

Answer: $\sqrt{x^2 + y^2}$ units

10. Show that $(3, 0)$, $(6, 4)$ and $(-1, 3)$ are the vertices of a right-angled isosceles triangle.

CBSE 2016, 2 marks

SOLUTION

Call the points $A(3, 0)$, $B(6, 4)$ and $C(-1, 3)$.

$$AB^2 = 3^2 + 4^2 = 25, AC^2 = (-4)^2 + 3^2 = 25 \text{ and } BC^2 = (-7)^2 + (-1)^2 = 50$$

$AB = AC$, so the triangle is isosceles. $AB^2 + AC^2 = 50 = BC^2$, so by the converse of the Pythagoras theorem $\angle A = 90^\circ$.

So the triangle is right-angled and isosceles. Proved.

11. The point $P(x, y)$ is equidistant from $A(a + b, b - a)$ and $B(a - b, a + b)$. Prove that $bx = ay$.

CBSE 2016, 3 marks

SOLUTION

$$PA^2 = PB^2: (x - a - b)^2 + (y - b + a)^2 = (x - a + b)^2 + (y - a - b)^2$$

$$\text{Use } p^2 - q^2 = (p - q)(p + q) \text{ on the } x \text{ terms: } (x - a - b)^2 - (x - a + b)^2 = (-2b)(2x - 2a) = -4bx + 4ab$$

$$\text{On the } y \text{ terms: } (y - b + a)^2 - (y - a - b)^2 = (2a)(2y - 2b) = 4ay - 4ab$$

$$\text{Adding the two results gives } 0: -4bx + 4ab + 4ay - 4ab = 0, \text{ so } 4ay = 4bx.$$

So $bx = ay$. Proved.

2 The section formula

12. Where does the mid-point of the segment joining $(5, -4)$ and $(6, 4)$ lie? (a) on the x -axis (b) on the y -axis (c) at the origin (d) on neither axis **CBSE 2026, 1 mark**

SOLUTION

$$\text{Mid-point} = \left(\frac{5+6}{2}, \frac{-4+4}{2} \right) = \left(\frac{11}{2}, 0 \right)$$

Its y -coordinate is 0 and its x -coordinate is not 0.

Answer: (a)

- 13.** In what ratio does the x -axis divide the segment joining $(-6, 5)$ and $(-4, -1)$? Find the point where they meet. **CBSE 2026, 3 marks**

SOLUTION

Let the ratio be $k : 1$. The point of division is $\left(\frac{-4k - 6}{k + 1}, \frac{-k + 5}{k + 1}\right)$.

On the x -axis the y -coordinate is 0: $-k + 5 = 0$, so $k = 5$.

$$x = \frac{-4(5) - 6}{5 + 1} = -\frac{26}{6} = -\frac{13}{3}$$

Answer: ratio 5 : 1; the point is $\left(-\frac{13}{3}, 0\right)$

- 14.** Where does the mid-point of the segment joining $P(-4, 5)$ and $Q(4, 6)$ lie? **CBSE 2025, 1 mark**

SOLUTION

$$\text{Mid-point} = \left(\frac{-4 + 4}{2}, \frac{5 + 6}{2}\right) = \left(0, \frac{11}{2}\right)$$

Its x -coordinate is 0.

Answer: on the y -axis

- 15.** In what ratio does the y -axis divide the segment joining $(5, -6)$ and $(-1, -4)$? Find the point where they meet. **CBSE 2025, 3 marks**

SOLUTION

Let the ratio be $k : 1$. The point of division is $\left(\frac{-k + 5}{k + 1}, \frac{-4k - 6}{k + 1}\right)$.

On the y -axis the x -coordinate is 0: $-k + 5 = 0$, so $k = 5$.

$$y = \frac{-4(5) - 6}{5 + 1} = -\frac{26}{6} = -\frac{13}{3}$$

Answer: ratio 5 : 1; the point is $\left(0, -\frac{13}{3}\right)$

- 16.** A circle has centre $(2, 0)$, and one end of a diameter is $(6, 0)$. Find the other end. (a) $(0, 0)$ (b) $(4, 0)$ (c) $(-2, 0)$ (d) $(-6, 0)$ **CBSE 2024, 1 mark**

SOLUTION

The centre is the mid-point of the diameter. Let the other end be (x, y) .

$$\frac{6 + x}{2} = 2 \text{ gives } x = -2, \text{ and } \frac{0 + y}{2} = 0 \text{ gives } y = 0.$$

Answer: (c)

- 17.** The point $\left(\frac{8}{5}, y\right)$ lies on the segment joining $(1, 2)$ and $(2, 3)$. In what ratio does it divide the segment? Find y as well. **CBSE 2024, 3 marks**

SOLUTION

Let the ratio be $k : 1$. Then $\frac{2k+1}{k+1} = \frac{8}{5}$.

$10k + 5 = 8k + 8$, so $k = \frac{3}{2}$. The ratio is $3 : 2$.

$$y = \frac{3 \times 3 + 2 \times 2}{3 + 2} = \frac{13}{5}$$

Answer: ratio $3 : 2$ and $y = \frac{13}{5}$

- 18.** $A(-1, -1)$, $B(-1, 6)$, $C(3, 6)$ and $D(3, -1)$ are the vertices of a rectangle. P , Q , R and S are the mid-points of AB , BC , CD and DA . Show that the diagonals of $PQRS$ bisect each other. **CBSE 2024, 3 marks**

SOLUTION

$$P = \left(-1, \frac{5}{2}\right), Q = (1, 6), R = \left(3, \frac{5}{2}\right), S = (1, -1)$$

$$\text{Mid-point of diagonal } PR = \left(\frac{-1+3}{2}, \frac{5/2+5/2}{2}\right) = \left(1, \frac{5}{2}\right)$$

$$\text{Mid-point of diagonal } QS = \left(\frac{1+1}{2}, \frac{6-1}{2}\right) = \left(1, \frac{5}{2}\right)$$

The diagonals PR and QS have the same mid-point, so they bisect each other.

- 19.** In what ratio does the x -axis divide the segment joining $A(3, 6)$ and $B(-12, -3)$? (a) $1 : 2$ (b) $1 : 4$ (c) $4 : 1$ (d) $2 : 1$ **CBSE 2023, 1 mark**

SOLUTION

Let the ratio be $k : 1$. On the x -axis the y -coordinate is 0: $\frac{-3k+6}{k+1} = 0$, so $k = 2$.

Answer: (d)

- 20.** AB is a diameter of a circle with centre $(2, -3)$, and B is the point $(1, 4)$. Find the coordinates of A . **CBSE 2019, 1 mark**

SOLUTION

The centre is the mid-point of the diameter. Let $A = (x, y)$.

$$\frac{x+1}{2} = 2 \text{ gives } x = 3, \text{ and } \frac{y+4}{2} = -3 \text{ gives } y = -10.$$

Answer: $A(3, -10)$

- 21.** In what ratio does the x -axis divide the segment joining $(1, -3)$ and $(4, 5)$? Find the point of division. **CBSE 2019, 2 marks**

SOLUTION

Let the ratio be $k : 1$. On the x -axis the y -coordinate is 0: $\frac{5k - 3}{k + 1} = 0$, so $k = \frac{3}{5}$.

The ratio is $3 : 5$. Then $x = \frac{3 \times 4 + 5 \times 1}{3 + 5} = \frac{17}{8}$.

Answer: ratio $3 : 5$; the point is $\left(\frac{17}{8}, 0\right)$

- 22.** The segment joining $A(2, 1)$ and $B(5, -8)$ is trisected at P and Q , with P nearer to A . P also lies on the line $2x - y + k = 0$. Find k . **CBSE 2019, 3 marks**

SOLUTION

P divides AB in the ratio $1 : 2$.

$$P = \left(\frac{1 \times 5 + 2 \times 2}{3}, \frac{1 \times (-8) + 2 \times 1}{3} \right) = (3, -2)$$

P lies on the line: $2(3) - (-2) + k = 0$, so $8 + k = 0$.

Answer: $k = -8$

- 23.** In what ratio does $P(4, m)$ divide the segment joining $A(2, 3)$ and $B(6, -3)$? Find m . **CBSE 2018, 2 marks**

SOLUTION

Let the ratio be $k : 1$. Compare x -coordinates: $\frac{6k + 2}{k + 1} = 4$, so $6k + 2 = 4k + 4$ and $k = 1$.

The ratio is $1 : 1$, so P is the mid-point: $m = \frac{3 + (-3)}{2} = 0$.

Answer: ratio $1 : 1$; $m = 0$

- 24.** $A(-2, 1)$, $B(a, 0)$, $C(4, b)$ and $D(1, 2)$ are the vertices of a parallelogram $ABCD$. Find a and b , and the lengths of its sides. **CBSE 2018, 3 marks**

SOLUTION

The diagonals of a parallelogram bisect each other, so AC and BD have the same mid-point.

$$\frac{-2 + 4}{2} = \frac{a + 1}{2} \text{ gives } a = 1, \text{ and } \frac{1 + b}{2} = \frac{0 + 2}{2} \text{ gives } b = 1.$$

$$AB = \sqrt{(1 + 2)^2 + (0 - 1)^2} = \sqrt{10} \text{ and } BC = \sqrt{(4 - 1)^2 + (1 - 0)^2} = \sqrt{10}$$

Opposite sides are equal, so $CD = \sqrt{10}$ and $DA = \sqrt{10}$.

Answer: $a = 1$, $b = 1$; each side is $\sqrt{10}$ units

- 25.** P and Q are the points of trisection of the segment joining $A(2, -2)$ and $B(-7, 4)$, with P nearer to A . Find P and Q . **CBSE 2016, 2 marks**

SOLUTION

$$P \text{ divides } AB \text{ in the ratio } 1 : 2: P = \left(\frac{-7 + 4}{3}, \frac{4 - 2}{3} \right) = (-1, 0)$$

$$Q \text{ divides } AB \text{ in the ratio } 2 : 1: Q = \left(\frac{-14 + 2}{3}, \frac{8 - 2}{3} \right) = (-4, 2)$$

Answer: $P(-1, 0)$ and $Q(-4, 2)$

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.